most important properties of tensor products are summarized in the following problem.

Problem 2.11.3. (a) Let U be any k-vector space. Construct a natural bijection between bilinear maps $V \times W \to U$ and linear maps $V \otimes W \to U$ ("natural" means that the bijection is defined without choosing bases).

(b) Show that if $\{v_i\}$ is a basis of V and $\{w_j\}$ is a basis of W, then $\{v_i \otimes w_j\}$ is a basis of $V \otimes W$.

(c) Construct a natural isomorphism $V^* \otimes W \to \mathrm{Hom}(V, W)$ in the case when V is finite dimensional.

(d) Let V be a vector space over a field k. Let $S^n V$ be the quotient of $V^{\otimes n}$ (n-fold tensor product of V) by the subspace spanned by the tensors $T - s(T)$ where $T \in V^{\otimes n}$ and s is a transposition. Also let $\wedge^n V$ be the quotient of $V^{\otimes n}$ by the subspace spanned by the tensors T such that $s(T) = T$ for some transposition s. These spaces are called the nth **symmetric power**, respectively **exterior power** of V. If $\{v_i\}$ is a basis of V, can you construct a basis of $S^n V, \wedge^n V$? If $\dim V = m$, what are their dimensions?

(e) If k has characteristic zero, find a natural identification of $S^n V$ with the space of $T \in V^{\otimes n}$ such that $T = sT$ for all transpositions s, and find a natural identification of $\wedge^n V$ with the space of $T \in V^{\otimes n}$ such that $T = -sT$ for all transpositions s.

(f) Let $A : V \to W$ be a linear operator. Then we have an operator $A^{\otimes n} : V^{\otimes n} \to W^{\otimes n}$ and its symmetric and exterior powers $S^n A : S^n V \to S^n W$, $\wedge^n A : \wedge^n V \to \wedge^n W$ which are defined in an obvious way. Suppose that $V = W$ and that $\dim V = N$, and that the eigenvalues of A are $\lambda_1, \ldots, \lambda_N$. Find $\mathrm{Tr}(S^n A)$ and $\mathrm{Tr}(\wedge^n A)$.

(g) Show that $\wedge^N A = \det(A)\mathrm{Id}$, and use this equality to give a one-line proof of the fact that $\det(AB) = \det(A)\det(B)$.

Remark 2.11.4. Note that a similar definition to the above can be used to define the tensor product $V \otimes_A W$, where A is any ring, V is a right A-module, and W is a left A-module. Namely, $V \otimes_A W$ is the abelian group which is the quotient of the group $V \bullet W$ freely generated by formal symbols $v \otimes w$, $v \in V$, $w \in W$, modulo the